DATA PREPARATION AND STATISTICAL ANALYSIS OF HOSPITAL PATIENT RECORDS

Transforming Raw Patient Data into Meaningful Insights for Better Healthcare Decisions

Engineer to Excel

1. PROBLEM STATEMENT

- Hospital patient records often contain missing values, duplicate entries, inconsistent formats and incorrect data.
- These data-quality issues can reduce the reliability of analysis and may lead to incorrect healthcare decisions.
- Therefore, a systematic approach is required to prepare, clean and analyze patient records and extract meaningful and reliable information.

2. OBJECTIVES

- Prepare and organize hospital patient records.
- Identify and handle missing values.
- Detect duplicate and incorrect records.
- Calculate important statistical measures.
- Visualize patient data using suitable charts.
- Identify patterns and variations in patient records.
- Generate meaningful insights for better healthcare decisions.

3. INTRODUCTION

- Hospital patient records contain valuable information such as age, gender, blood pressure, heart rate, diagnosis, admission type and length of hospital stay.
- Data preparation converts raw records into a clean and consistent form suitable for analysis.
- Statistical analysis helps summarize patient characteristics, identify trends and understand relationships within the data.
- These insights can support better patient care, hospital management and data-driven decision-making.

4. DATASET (Sample Patient Records)

Patient ID	Age (Years	Gender	BP (mmH	Heart Rate (bpm)	Stay (Days)	Diagnosis
P001	45	Male	130	82	5	Hypertension
P002	32	Female	118	76	3	Asthma
P003	67	Male	145	90	8	Diabetes
P004	51	Female	125	80	4	Hypertension
P005	39	Male	120	74	2	Fever
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Variables:

- Age – Age of the patient (years)
- Gender – Male (M) / Female (F)
- BP – Blood Pressure (mmHg)
- Heart Rate – beats per minute
- Stay – Hospital stay duration (days)
- Diagnosis – Medical condition

CHALLENGES:

Hospital datasets may contain several issues that can affect the accuracy of analysis:
Missing Data: Some patient records may have blank values.
Duplicate Records: The same patient may appear more than once.
Inconsistent Data: Values may be recorded in different formats.

5. METHOD PIPELINE

1. Patient Records (Raw Data) → **2. Data Cleaning** → **3. Missing Value Handling** → **4. Data Transformation** → **5.Statistical Analysis** → **6. Visualization** → **7. Healthcare Insights**

1. Patient Records (Raw Data): Collected from hospital systems
2. Data Cleaning: Handle missing values, duplicates and errors
3. Missing Value Handling: Impute using Mean / Median / Mode
4. Data Transformation: Standardize formats, encode categorical
5. Statistical Analysis: Compute measures and summary statistics
6. Visualization: Create charts and graphs to explore patterns
7. Healthcare Insights: Generate meaningful insights for decisions

6. STATISTICAL ANALYSIS

MEAN (μ) Average of all values.
$\mu = (\Sigma x) / n$

MEDIAN Middle value when data is arranged in ascending order.

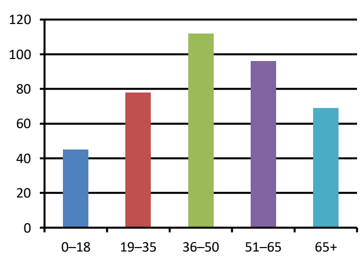
MODE Value that occurs most frequently.

RANGE Difference between maximum and minimum values.
Range = Max − Min

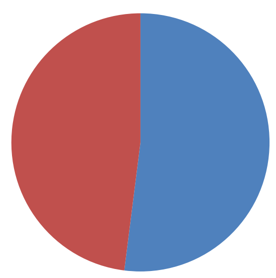
STANDARD DEVIATION (σ) Measures the spread of data around the mean.
$\sigma = \sqrt{[\Sigma(x_i - \mu)^2 / n]}$

7. RESULTS & VISUALIZATION

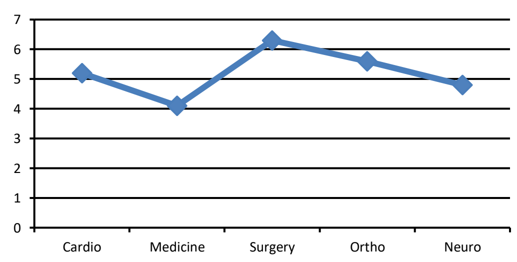
Patients by Age Group



Gender Distribution



Average Hospital Stay (Days)



Average BP: 128.6 / 82.4 mmHg

Top Diagnosis: Hypertension

Highest Average Stay: Surgery

8. CONCLUSION

Proper data preparation plays a vital role in improving the quality and reliability of hospital patient records. Statistical analysis helps identify important patterns, trends and variations in patient data.

These insights support healthcare professionals in making data-driven decisions, improving patient care, optimizing resources and enhancing overall hospital performance.

9. FUTURE SCOPE

- Apply machine learning models for disease prediction.
- Predict patient readmission and risk of complications.
- Integrate real-time data from IoT medical devices.
- Use advanced analytics for personalized treatment plans.
- Develop dashboards for real-time hospital monitoring and decision support.

SLOT-A
DSA0404-Fundamentals of Data Science

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